Coding & Solution

Date	29 June 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/JohnsonpaulG/Credit-Card-Approval-Prediction
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy
Key Features Implemented	Credit card approval prediction, applicant data preprocessing, machine learning prediction, input validation, responsive web interface, real-time prediction results
Pending / Incomplete Features	User authentication, database integration, prediction history, cloud deployment, email notifications
Setup / Run Instructions	Clone the project repository, install dependencies using **pip install -r requirements.txt**, then run the application using **python app.py**. Open the local URL in a browser to use the application.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The application follows a modular architecture by separating the frontend, backend, and Machine Learning components.
- Applicant information is pre-processed before being passed to the trained Machine Learning model for prediction.
- Flask is used to connect the user interface with the prediction model and display results in real time.
- The project is designed to support future enhancements such as user authentication, database integration, cloud deployment, and additional Machine Learning models.
- The code is organized to improve readability, maintainability, and reusability.